Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/05/2005
PI Name:	Loll, Patrick J. Ph.D.		
Project Title:	Detergent Interactions Affecting Membrane Protein Crystallization: Analysis and Use in Screen Design		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19102-1192	**Congressional District:**	2
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	09/01/2001	**End Date:**	11/30/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wiencek, J. (University of Iowa)		
Grant/Contract No.:	NAG8-1840		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The quantitative analysis of detergent-detergent interactions will be carried out using measurements of the second osmotic virial coefficient (B22) obtained from static light scattering using the reaction center protein from Rb. sphaeroides. In addition, dynamic light scattering, infrared spectroscopic imaging, specific volume measurements, and solubility studies will be used to extend our understanding of the crystallization behavior of OmpF porin. To test our crystallization screens, we will express, purify, and study the crystallization behavior of four to six new (as yet uncrystallized) membrane proteins. These proteins will be analyzed in at least 12 different detergents, under conditions previously identified as promising for crystal formation. The principal objectives are to extend our quantitative analysis of detergent-detergent interactions and their contribution to crystal nucleation and growth, and to test experimental crystallization screens that we have recently developed using a battery of hitherto uncrystallized membrane proteins. Protein structural information is critical for structure-based drug design, protein engineering, and basic studies of how living cells function. The most powerful technique to obtain this knowledge is x-ray crystallography; this method,		

	however, has been nearly useless for membrane proteins, since this class of molecules has proven extremely difficult to crystallize. We are developing rational methods to overcome these difficulties, based upon the manipulation of the interaction forces underlying the crystallization behavior of complexes of membrane proteins and surfactant. We expect that this work will place membrane protein crystallization on a more rational footing, and believe that once the crystallization bottleneck is removed, we will be able to determine the atomic structures of these extremely important molecules in a straight-forward manner.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the last fiscal year, we have been focusing on preparing proteins for use in testing our crystallization screens. Expression constructs have been prepared for 4 members of the small multidrug resistance transporter family from 4 different bacterial species, and expression has been verified by Western blotting. Levels are low, suggesting that obtaining enough material for exhaustive testing of screens may prove problematic. We have also produced good expression systems for outer membrane glycosyl transferase enzymes from three different bacterial species that produce yields in excess of 5 mg/liter of culture. Optimization of the purification conditions is currently underway for these outer membrane proteins.
Bibliography Type:	Description: (Last Updated: 01/05/2005)
Abstracts for Journals and Proceedings	Zakharov, S.D.; Eroukova, V.Y.; Rokitskaya, T.I.; Zhalnina, M.V.; Sharma, O.; Loll, P.J.; Zgurskaya, H.I.; Antonenko, Y.N.; Cramer, W.A. "Colicin occlusion of OmpF and TolC channels: outer membrane translocons for colicin import" Biophysical Journal Vol. 87, pp. 3901-3911 , Dec-2004